

A Sophie-Prime Reduction for Erdős Problem #287

A conditional generated-Type-II criterion with a Lean-verified finite obstruction

Public review draft v0.2

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Claim level. This is a public review draft, not a solution claim. The finite Erdős–287 obstruction layer is machine-checked in Lean, including the exact public-problem bridge, sign-sensitive Sophie blockers, and the exclusion of exact counterexamples with maximum denominator at most $4 \cdot 10^9$. The paper-level comparison, Type-I and rough upper-sieve steps are reduced to classical cited results. The central parity-breaking input—a generated-coefficient Type-II estimate, together with its packet census and Ford–Maynard replacement certificate—is **not proved**. No solution of Erdős Problem #287, the Twin Prime Conjecture, or the Sophie Germain prime conjecture is claimed.

Version note. This v0.2 supersedes the earlier public draft in three respects: (i) the physical Erdős–287 affine shell is $2C_J a_J - dpl = -s$ ($s = \pm 1$), not the twin-shift shell $C_J a_J - dpl = -2$; (ii) the balanced nine-prime source is packetised only after a smooth product cutoff and a finite logarithmic-polytope partition; and (iii) the accompanying Lean development now verifies the exact public-problem bridge and a finite range through $4 \cdot 10^9$.

Abstract

Erdős Problem #287 asks whether every finite set of distinct integers greater than one whose reciprocals sum to one has a consecutive gap at least three. We give a short self-contained finite reduction: a top-layer p -adic congruence forces certain prime multiples to be absent, and a prime q in a fixed compact subinterval of $(2X/3, X)$ for which one of $2q - 1, 2q + 1$ is prime produces two adjacent holes. A Lean development formalises the exact public counterexample predicate, the finite obstruction, and a certificate mechanism excluding maximum denominators up to $4 \cdot 10^9$ without user axioms or **sorry**. We then record the exact comparison density for the two affine forms, a mod-6 source split, a classical Type-I reduction, and a rough two-linear-form upper-sieve reduction. The remaining nonstandard input is isolated as a generated-coefficient Type-II estimate for the finite family of coefficient pairs actually produced in a fixed Ford–Maynard rerun. If that estimate and its source certificate are supplied, the Ford–Maynard lower-bound argument yields a Sophie-type witness for every sufficiently large scale; an effective finite remainder would then complete the problem.

1 Finite reduction from the reciprocal identity

Let

$$A = \{n_1 < \cdots < n_k\} \subset \mathbb{N}, \quad n_1 > 1, \quad \sum_{a \in A} \frac{1}{a} = 1.$$

Write $N = n_1$ and $M = n_k$, and suppose for contradiction that every consecutive gap is at most 2. Call $h \in [N, M]$ a *hole* if $h \notin A$.

Lemma 1.1 (Adjacent holes). *There are no two adjacent holes in $[N, M]$.*

Proof. If $x, x + 1$ were holes, the closest members of A on their two sides would differ by at least 3. $\square$

Lemma 1.2 (Half-range placement). *Every putative counterexample satisfies $N \leq \lfloor M/2 \rfloor$.*

Proof. If $N \geq \lfloor M/2 \rfloor + 1$, then $k \leq M - N + 1 \leq N$. Since the denominators are distinct and at least N ,

$$\sum_{i=1}^k \frac{1}{n_i} \leq \frac{1}{N} + \frac{k-1}{N+1} < \frac{k}{N} \leq 1,$$

a contradiction. The case $k = 1$ would force $n_1 = 1$ and is excluded. $\square$

Theorem 1.3 (Top-layer congruence). *Fix a prime p , put $E = \max_{a \in A} v_p(a)$ and $I_p = \{a \in A : v_p(a) = E\}$. If $a = p^E m_a$ with $p \nmid m_a$, then*

$$\sum_{a \in I_p} m_a^{-1} \equiv 0 \pmod{p}.$$

Proof. Write $L = \text{lcm}(A) = p^E L_0$ with $p \nmid L_0$. Multiply the reciprocal identity by L and reduce modulo p . Terms of smaller p -adic valuation vanish; the top-layer terms are $L_0 m_a^{-1}$. Since L_0 is a unit, the congruence follows. $\square$

Lemma 1.4 (Prime-fibre exclusions). *Let $q > 3$ be prime with $3q > M$ and $q^2 > M$. Then neither q nor $2q$ lies in A . Every prime $p \in (M/2, M]$ is also absent from A .*

Proof. The only possible multiples of q are $q, 2q$, both of q -adic valuation one. The possible nonempty top fibres give $1, 2^{-1}$, or $1 + 2^{-1}$ modulo q , none zero for $q > 3$. For $p > M/2$ the top fibre would be the singleton $\{p\}$. $\square$

2 Sophie-type witnesses and the formal finite range

Put $X = \lfloor M/2 \rfloor$.

Proposition 2.1 (Sign-sensitive witness). *If there is a prime $q \geq 5$ with $M < 3q$ and either*

$$2q + 1 \leq M, \quad 2q + 1 \text{ prime},$$

or

$$2q \leq M, \quad 2q - 1 \text{ prime},$$

then no gap- ≤ 2 counterexample with maximum M exists.

Proof. The previous lemma makes $2q$ a hole. In the plus case $2q + 1$ is a top-half prime and is the adjacent hole; in the minus case $2q - 1$ is a top-half prime and is the adjacent hole on the other side. The half-range lemma places both holes above N . $\square$

Equivalently, the plus branch allows $M/3 < q \leq (M - 1)/2$, while the minus branch allows the slightly wider $M/3 < q \leq \lfloor M/2 \rfloor$.

Theorem 2.2 (Asymptotic fixed-band reduction). *Let $J \in (2/3, 1)$ be fixed. For all sufficiently large M , if there is a prime q with $q/X \in J$ and at least one of $2q - 1, 2q + 1$ prime, then no counterexample with maximum M exists.*

The concrete support $J = (3/4, 4/5)$ already works for $M \geq 9$.

Lean-verified finite statement

The accompanying Lean development defines an exact predicate `Erdos287Counterexample` matching the public problem and proves the one-way bridge to the historical finite compiler. It proves the sign-sensitive witness theorem above and a window-certificate engine based on prime-power divisors of two consecutive integers. A chain of 34 kernel-checked interval certificates proves

$$\boxed{\text{No exact counterexample has } 3 \leq M \leq 4,000,000,000.}$$

The project builds without `sorry`, `admit`, user `axiom`, `unsafe`, or `native_decide`; the principal theorem `axiom` printouts contain only the usual Lean foundations `propext`, `Classical.choice`, and `Quot.sound`. No unconditional theorem named `erdos287` is declared.

3 The exact comparison density and a mod-6 split

Let $W \geq 0$ be smooth and supported in $J \in (2/3, 1)$. Define

$$\mathfrak{S}_2 = \prod_{p>2} \frac{p(p-2)}{(p-1)^2}, \quad B(n) = \mathfrak{S}_2 \prod_{\substack{p|n \\ p>2}} \frac{p-1}{p-2},$$

$$a_X(n) = W(n/X)(\Lambda(2n-1) + \Lambda(2n+1)), \quad b_X(n) = 4W(n/X)B(n), \quad w_X = a_X - b_X.$$

Proposition 3.1 (Local mean). *For every fixed positive integer d ,*

$$\lim_{Y \rightarrow \infty} \frac{1}{Y} \sum_{m \leq Y} B(dm) = \frac{d}{\varphi(2d)}.$$

Proof. For odd squarefree e put $g(e) = \prod_{p|e} (p-2)^{-1}$ and put $g(e) = 0$ otherwise. Then $B(n) = \mathfrak{S}_2 \sum_{e|n} g(e)$ and $\sum_e |g(e)|/e < \infty$. Averaging produces $\mathfrak{S}_2 \sum_e g(e)(e, d)/e$. Its Euler factor is 1 for $p \nmid d$ and $p/(p-1)$ for $p \mid d$. $\square$

The mod-6 sifted source

$$a_X^{(6)} = \mathbf{1}_{(n,6)=1} a_X,$$

splits, up to total von Mangoldt mass $O(\log X)$ from powers of 3, into two disjoint lanes

$$n \equiv 1 \pmod{6} : (n, 2n-1), \quad n \equiv 5 \pmod{6} : (n, 2n+1).$$

This is the preferred source for any later Type-II analysis because it removes sign mixing without changing the prime conclusion.

4 Standard analytic inputs outside the parity barrier

The point of this section is not novelty; it is to isolate the one nonstandard estimate.

Proposition 4.1 (Comparison and growth). *The sequence b_X has prime mass $\asymp X/\log X$ and satisfies the Ford–Maynard comparison condition on each fixed-dimensional rough cell after standard multivariate prime-number-theorem partial summation. Moreover*

$$B(n) \ll \log \log(3n), \quad \sum_n |w_X(n)| \tau(n) \ll_W X(\log X)^2.$$

Proposition 4.2 (Type I: classical reduction). *For each fixed $\varepsilon > 0$ and $A > 0$,*

$$\sum_{m \leq X^{1/2-\varepsilon}} \tau(m)^A \max_I \left| \sum_{\substack{X/2 < mn \leq X \\ n \in I}} w_X(mn) \right| \ll_{A,W,\varepsilon} X(\log X)^{-A}.$$

Reduction. For fixed m , the two prime terms are prime-number-theorem remainders in the reduced classes $\pm 1 \pmod{2m}$. Partial summation inserts W , and the maximal Bombieri–Vinogradov theorem gives arbitrary logarithmic saving uniformly up to $2m \ll X^{1/2-\varepsilon}$. A standard divisor-moment truncation absorbs $\tau(m)^A$. The comparison term is evaluated in progressions from the absolutely convergent divisor expansion of B . $\square$

Proposition 4.3 (Rough exceptional collar: upper-sieve reduction). *At a fixed $0 < \varepsilon \leq 1/600$, put $\sigma = 1/6 - 2\varepsilon$. On the σ -rough exceptional Ford–Maynard sector, squareful terms are negligible and $\Omega(n) \leq 6$. After choosing the collar prime q , write $n = qdr$ with r another prime. A dimension-two Selberg upper sieve applied to*

$$r, \quad 2qdr \pm 1$$

gives a uniform bound of the required $X/\log X$ scale, with local correction

$$\prod_{\ell|qd} \left(1 - \frac{1}{\ell}\right)^{-1} = 1 + o(1)$$

on the rough support.

These reductions use classical maximal Bombieri–Vinogradov and fixed-dimensional Selberg upper-sieve technology; they are not the parity-breaking step.

5 The exact central open theorem

Ford and Maynard develop a general prime-producing sieve for arbitrary nonnegative sequences satisfying Type I and Type II estimates. Their proof of Proposition 7.22 reaches a grouped bilinear expression (their equation (7.23)) and invokes Type II only after forming the generated coefficient pair. This motivates the following proof-specific target.

Hypothesis 5.1 (Generated Erdős–287 Type II). *There is one fixed $0 < \varepsilon_* \leq 1/600$ such that every grouped coefficient pair (Y_1, Y_2) actually generated in a fixed Ford–Maynard rerun satisfies, for every $A > 0$,*

$$\sum_{\substack{(X/2)^{\varepsilon_*} < m \leq X^{1/6-\varepsilon_*} \\ X/2 < mn \leq X}} Y_1(m)Y_2(n)W(mn/X)[\Lambda(2mn-1) + \Lambda(2mn+1) - 4B(mn)] \ll_A X(\log X)^{-A}. \quad (\text{TII-287})$$

This must be accompanied by (i) a finite census of every generated equation-(7.23) packet, including ranges, coefficient norms, multiplicity and Mellin/Perron costs, and (ii) an invocation-by-invocation replacement certificate C_{FM} showing that every Type-II use in the chosen Ford–Maynard rerun is one of the proved packets. A theorem for a merely similar coefficient family is not enough.

Theorem 5.2 (Conditional Erdős–287 criterion). *Assume (TII-287), the complete packet census and C_{FM} , and choose ε_* so that the Ford–Maynard positive central margin dominates the $O(\varepsilon_*)$ rough exceptional loss. Then*

$$\sum_{q \text{ prime}} W(q/X)(\Lambda(2q-1) + \Lambda(2q+1)) \gg_W \frac{X}{\log X}.$$

Hence no sufficiently large counterexample to Erdős Problem #287 exists. A complete solution would additionally require effective constants and a finite certificate below the resulting threshold.

6 Affine correction and relation to the separate twin-prime programme

The technical programme that led to the present Type-II architecture began with a different physical source, a single fixed twin shift such as $uv + 2 = q\ell$. That source is *not* the Erdős–287 source. For one 287 sign $s \in \{\pm 1\}$ the literal complementary-divisor equation is

$$2uv + s = q\ell,$$

and, after a nine-prime defect split, the physical shell is

$$\boxed{2C_J a_J - dp\ell = -s.}$$

In particular, the earlier shell $C_J a_J - dp\ell = -2$ must not be used as a 287 packet.

The finite front algebra does transfer after the constants are changed. If $Auv + B = q\ell$, then

$$(u, \ell) \mid B, \quad (v, q) \mid B,$$

and two incidences satisfy

$$v_2 q_1 \ell_1 - v_1 q_2 \ell_2 = B(v_2 - v_1).$$

For the 287 lanes $(A, B) = (2, \pm 1)$ the gcds are therefore trivial. If q is odd and A is a unit modulo q , the complete Fourier kernel is the standard Kloosterman sum

$$\sum_{\substack{x, y \bmod q \\ Axy \equiv -s \pmod{q}}} e_q(hx + ky) = S(h, -sk\bar{A}; q).$$

What does *not* automatically transfer is the physical Poisson/Cauchy/ TT^* weight, its ranges, multiplicity and source norms. Those must be rederived from $2uv \pm 1 = q\ell$ before any fixed-shift Gate theorem can be used as a provider for (TII-287).

Twin-prime relevance / provenance. The separate fixed-shift programme contains formal algebra and candidate analytic machinery that may provide technology for the generated Type-II estimate above. It is being released separately as a public research snapshot so that its exact ideas and status are independently inspectable. This paper does not claim that the fixed-shift programme proves (TII-287), and it makes no claim that the Twin Prime Conjecture is solved.

7 Formal verification status

The accompanying Lean repository proves, without user axioms or `sorry`:

- an exact public counterexample predicate and the one-way bridge to the historical `Gap2CE` compiler;
- the top-layer, half-range, sign-sensitive Sophie and maximum-divisor blockers;
- the interval-certificate engine and the exclusion $3 \leq M \leq 4 \cdot 10^9$;
- deterministic conditional compilers whose hypotheses remain explicit and uninhabited.

It does *not* formalise the analytic Type-II hypothesis, the Ford–Maynard lower-bound rerun, or the final theorem #287. This scope distinction is intentional.

References

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